# End-Term Sample Question Paper - ANSWER KEY

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set B - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

*Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.*

*For the instructor. Every numerical result in this key was recomputed and verified before printing; the marks in brackets give the part-wise marking scheme.*

## Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

**Q1.** Define an equivalence relation, name its three defining properties, and state how many equivalence classes congruence modulo 5 produces on $\mathbb{Z}$. **(2 Marks)**

*[Unit I · CO1 · L1 - Remember · Theory]*

An equivalence relation is **reflexive, symmetric and transitive**. (1 mark)

Congruence modulo 5 gives **5** classes: [0], [1], [2], [3], [4]. (1 mark)

**Q2.** For A = [[6, 3], [4, 5]], compute det A and state whether A is invertible, with the reason. **(2 Marks)**

*[Unit II · CO2 · L2 - Understand · Numerical]*

$\det A = 6(5) - 3(4) = 30 - 12 = \mathbf{18}$. (1 mark)

$\det A \neq 0$, so A **is invertible**. (1 mark)

**Q3.** State the mean and the variance of a binomial random variable with n = 25 and p = 0.4. **(2 Marks)**

*[Unit III · CO3 · L2 - Understand · Numerical]*

Mean = np = 25(0.4) = **10**. (1 mark)

Variance = np(1 − p) = 25(0.4)(0.6) = **6**. (1 mark)

**Q4.** State the chromatic number of the cycle $C_6$, the cycle $C_7$ and the complete graph $K_5$, with a one-line reason for the last. **(2 Marks)**

*[Unit IV · CO4 · L1 - Remember · Theory]*

$\chi(C_6) = \mathbf{2}$ (even cycle), $\chi(C_7) = \mathbf{3}$ (odd cycle). (1 mark)

$\chi(K_5) = \mathbf{5}$: every vertex is adjacent to all four others, so no two may share a colour. (1 mark)

**Q5.** State the three cases of the Master theorem for T(n) = aT(n/b) + f(n), and give the critical exponent that the cases compare f(n) against. **(2 Marks)**

*[Unit V · CO5 · L2 - Understand · Theory]*

The critical exponent is **n^(log_b a)**. (0.5 marks)

**Case 1:** f(n) polynomially smaller → Θ(n^(log_b a)). **Case 2:** f(n) = Θ(n^(log_b a)) → Θ(n^(log_b a) log n). **Case 3:** f(n) polynomially larger and regular → Θ(f(n)). (1.5 marks)

## Part B - Short Answer Questions (4 x 5 = 20 Marks)

**Q6.** Build the Huffman code for the symbol frequencies A = 2, B = 3, C = 4, D = 5, E = 6. List the merges in order and compute the total number of encoded bits. **(5 Marks)**

*[Unit VI · CO6 · L3 - Apply · Numerical]*

Merges: 2 + 3 = **5**; 4 + 5 = **9**; 5 + 6 = **11**; 9 + 11 = **20**. (2 marks)

Code lengths: A and B get 4 bits each (deepest, merged first); C and D get 2 bits; E gets 2 bits. Any tree produced by the algorithm is acceptable provided the total is optimal. (1 mark)

Total bits = sum of the internal-node weights = 5 + 9 + 11 + 20 = **45 bits**. (2 marks)

**Q7.** Find the multiplicative inverse of 9 modulo 31, and use it to solve $9x \equiv 5 \pmod{31}$ for the least non-negative x. State why the inverse exists. **(5 Marks)**

*[Unit I · CO1 · L3 - Apply · Numerical]*

gcd(9, 31) = 1 (31 is prime and does not divide 9), so the inverse exists. (1 mark)

$9(7) = 63 = 2(31) + 1$, so $\mathbf{9^{-1} \equiv 7 \pmod{31}}$. (2 marks)

$x \equiv 7(5) = 35 \equiv 35 - 31 = \mathbf{4 \pmod{31}}$. (1.5 marks)

Check: $9(4) = 36 = 31 + 5$, so $9(4) \equiv 5 \pmod{31}$. Correct. (0.5 marks)

**Q8.** Distinguish a discrete from a continuous random variable, define the expectation and the variance of a discrete random variable, and state what the variance measures. **(5 Marks)**

*[Unit III · CO3 · L2 - Understand · Theory]*

A **discrete** random variable takes countably many values with a probability mass at each; a **continuous** one takes values in an interval, where any single value has probability 0 and probabilities come from areas under a density. (2 marks)

$E[X] = \Sigma\, x_i\, p_i$; $\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \Sigma\, (x_i - \mu)^2\, p_i$. (2 marks)

The variance measures **spread about the mean**; its square root, the standard deviation, is in the same units as X. (1 mark)

**Q9.** Define the classes P, NP, NP-hard and NP-complete, and explain in two sentences what a polynomial-time reduction is used for. **(5 Marks)**

*[Unit V · CO5 · L4 - Analyse · Theory]*

**P:** solvable in polynomial time. **NP:** a proposed solution can be **verified** in polynomial time. (2 marks)

**NP-hard:** at least as hard as every problem in NP. **NP-complete:** in NP **and** NP-hard — the hardest problems inside NP. (2 marks)

A reduction transforms instances of X into instances of Y in polynomial time, so an efficient algorithm for Y yields one for X. It therefore transfers **hardness** in the opposite direction, which is how NP-completeness is proved. (1 mark)

## Part C - Long Answer Questions (3 x 10 = 30 Marks)

**Q10.** Eigenvalues and eigenvectors. **(10 Marks)**

*[Unit II · CO2 · L4 - Analyse · Theory + Numerical]*

(a) Define an eigenvalue and an eigenvector, state the characteristic equation, and state what the spectral theorem guarantees for a real symmetric matrix. *[5]*

(b) For A = [[7, 2], [2, 4]], find both eigenvalues and one eigenvector for each, and verify your eigenvalues using the trace and the determinant. *[5]*

**(a)** A non-zero vector v is an **eigenvector** of A with **eigenvalue** $\lambda$ if $Av = \lambda v$. The characteristic equation is $\det(A - \lambda I) = 0$. (3 marks)

**Spectral theorem:** a real symmetric matrix has all-**real** eigenvalues and a set of **mutually orthogonal** eigenvectors — which is what makes the principal components of a covariance matrix a genuine orthonormal basis. (2 marks)

**(b)** $\mathrm{tr}\, A = 11$, $\det A = 28 - 4 = 24$, so $\lambda^2 - 11\lambda + 24 = 0$, i.e. $(\lambda - 8)(\lambda - 3) = 0$: $\mathbf{\lambda_1 = 8, \lambda_2 = 3}$. (2 marks)

$\lambda = 8$: $(A - 8I)v = [[-1, 2], [2, -4]]v = 0$ gives $v_1 = 2v_2$, so $\mathbf{v_1 = (2, 1)}$. (1.5 marks)

$\lambda = 3$: $(A - 3I)v = [[4, 2], [2, 1]]v = 0$ gives $2v_1 + v_2 = 0$, so $\mathbf{v_2 = (1, -2)}$. (1 mark)

Check: $8 + 3 = 11 = \mathrm{tr}\, A$ and $8 \times 3 = 24 = \det A$. Also $v_1 \cdot v_2 = 2 - 2 = 0$, orthogonal as the spectral theorem requires. (0.5 marks)

**Q11.** Minimum spanning trees. **(10 Marks)**

*[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]*

(a) State what a minimum spanning tree is, how many edges it has on n vertices, and describe how Kruskal's algorithm builds one. *[5]*

(b) Apply Kruskal to the undirected graph with AB = 4, AC = 1, BC = 2, BD = 5, CD = 8, DE = 3, CE = 9. List every edge accepted and every edge rejected, and give the total weight. *[5]*

**(a)** An MST is an acyclic connected subgraph containing every vertex, of least total edge weight. On n vertices it has exactly **n − 1** edges. (2 marks)

Kruskal sorts all edges by weight and adds each in turn, skipping any edge whose endpoints are already connected (which would close a cycle), until n − 1 edges are chosen. A disjoint-set structure performs the connectivity test. (3 marks)

**(b)** Sorted: AC(1), BC(2), DE(3), AB(4), BD(5), CD(8), CE(9). (1 mark)

Accept **AC(1)**; accept **BC(2)**; accept **DE(3)**; **reject AB(4)** — A and B already joined through C; accept **BD(5)**, which links {A,B,C} to {D,E}. (2.5 marks)

Four edges chosen for five vertices, so stop; CD and CE are never examined. (0.5 marks)

Total weight = 1 + 2 + 3 + 5 = **11**. (1 mark)

**Q12.** Dynamic programming for the 0/1 knapsack. **(10 Marks)**

*[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]*

(a) State the DP recurrence for V[i, w], and explain which two properties a problem must have before dynamic programming is worth using. *[5]*

(b) Build the complete table for weights {2, 3, 4}, values {25, 30, 50} and capacity W = 6, state the optimal value, and trace back the chosen items. *[5]*

**(a)** V[i, w] = V[i−1, w] if $w < w_i$, otherwise max(V[i−1, w], V[i−1, $w - w_i$] + $v_i$). (2.5 marks)

It needs **optimal substructure** and **overlapping subproblems**. Optimal substructure alone suits divide and conquer; it is the overlap that makes storing subproblem answers pay. (2.5 marks)

**(b)** Table (rows i = 0..3, columns w = 0..6):

i=0: 0 0 0 0 0 0 0 i=1 (2,25): 0 0 25 25 25 25 25

i=2 (3,30): 0 0 25 30 30 55 55 i=3 (4,50): 0 0 25 30 50 55 **75** (2 marks)

Optimal value = **75**. (1 mark)

Traceback: V[3,6] = 75 ≠ V[2,6] = 55, so item 3 is taken (4 kg), leaving capacity 2; V[2,2] = 25 = V[1,2], so item 2 is not taken; V[1,2] = 25 ≠ V[0,2] = 0, so item 1 is taken. Chosen **{item 1, item 3}**, weight 2 + 4 = 6, value 25 + 50 = 75. (2 marks)

**----- End of Answer Key -----**